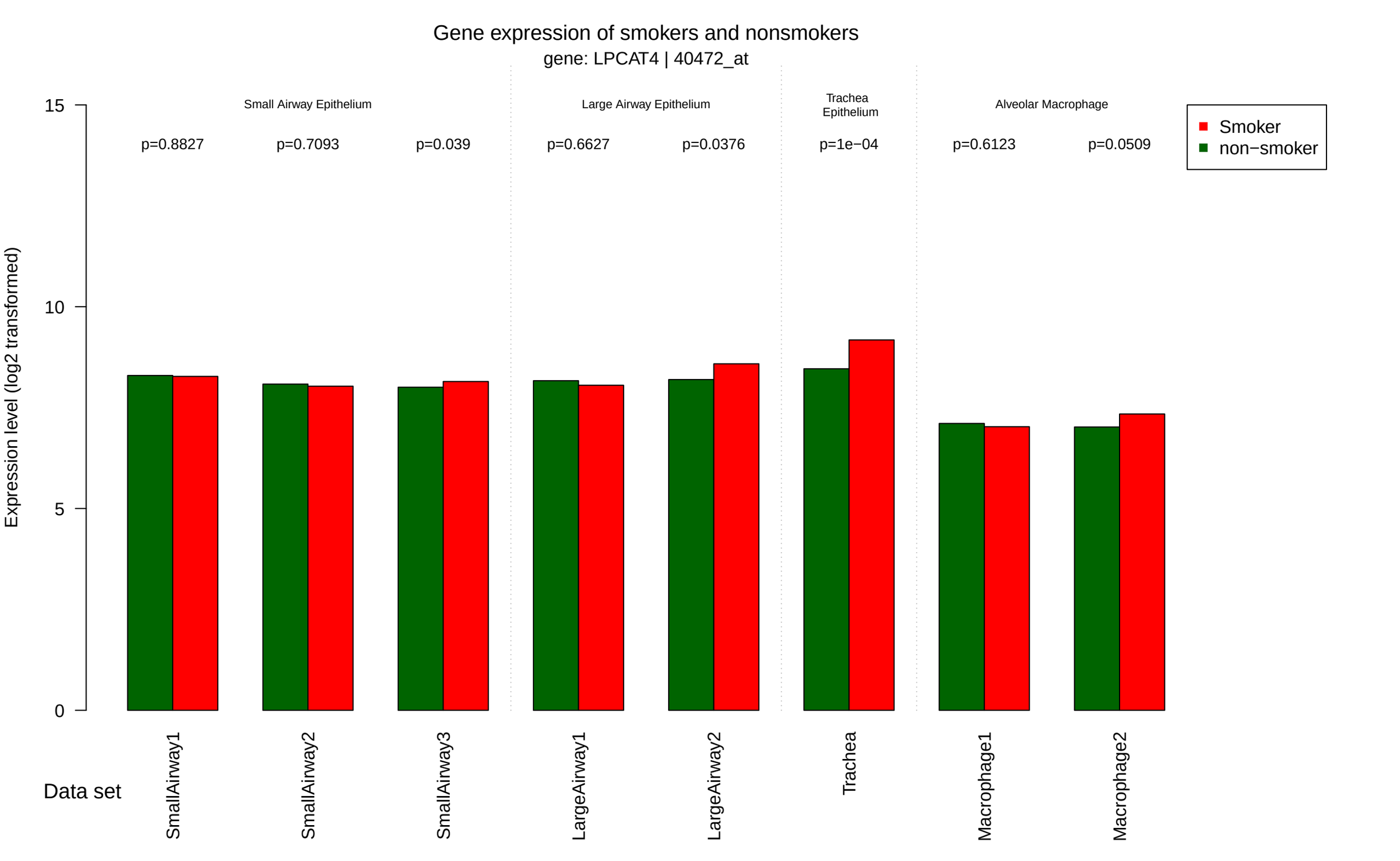
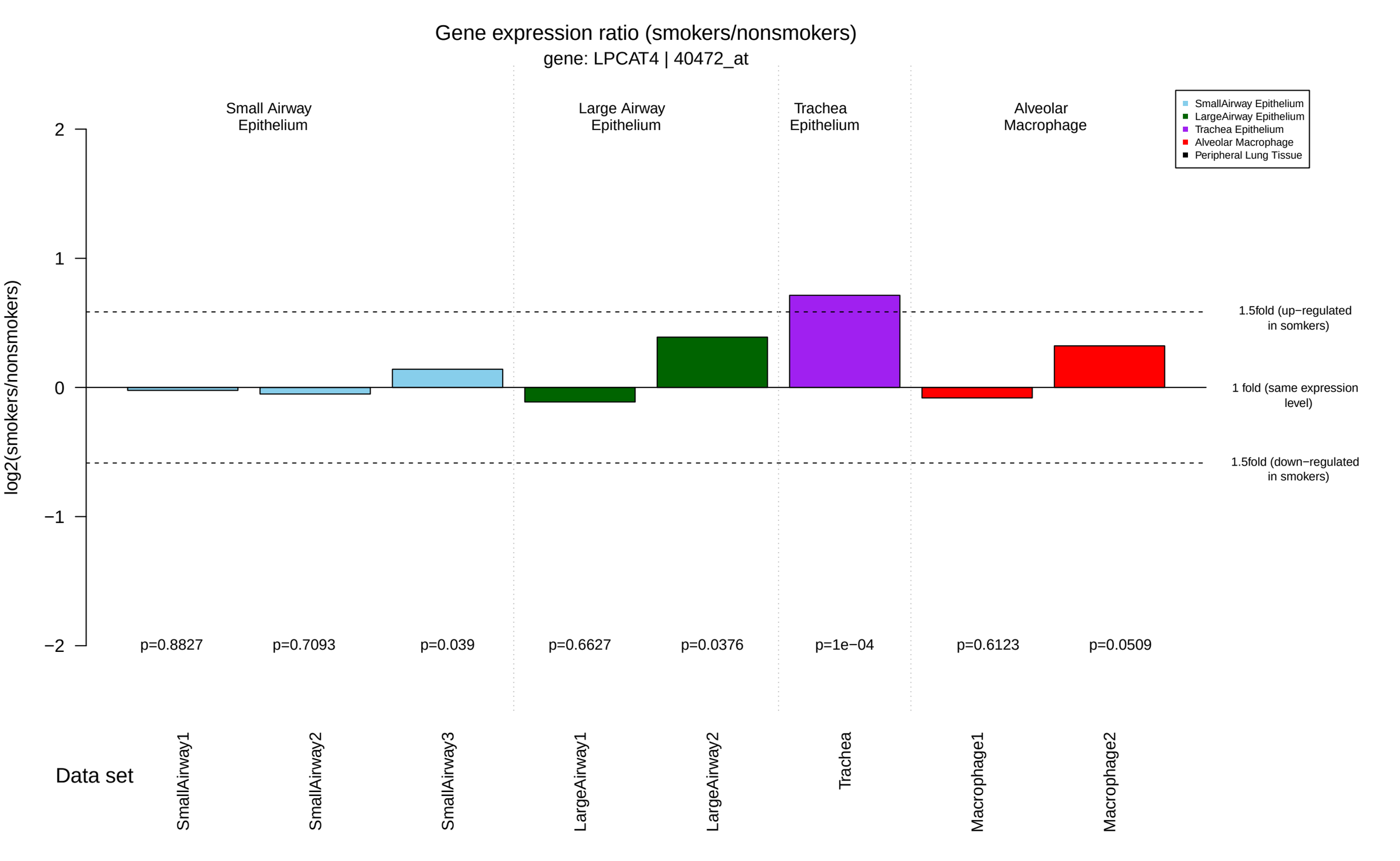
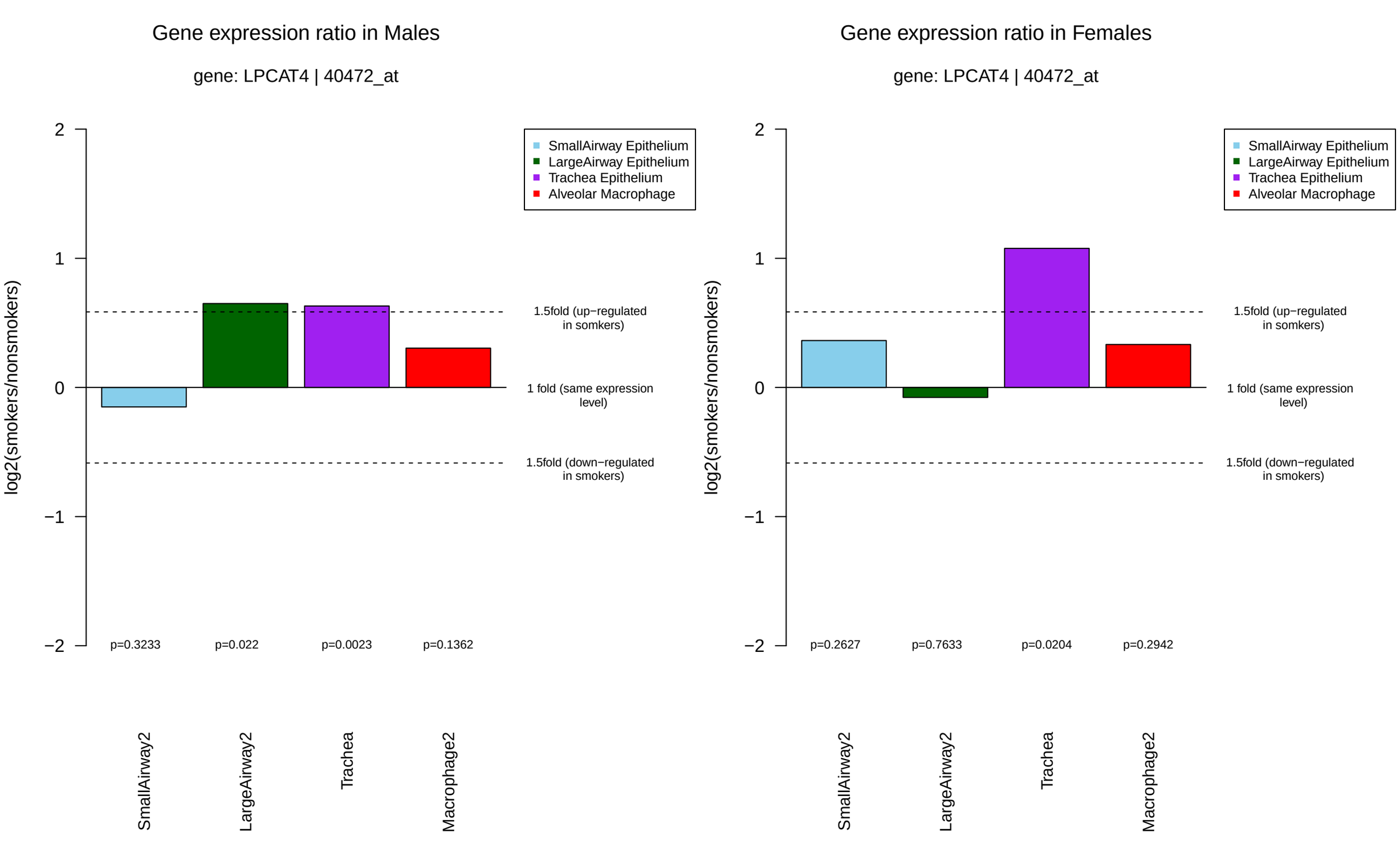
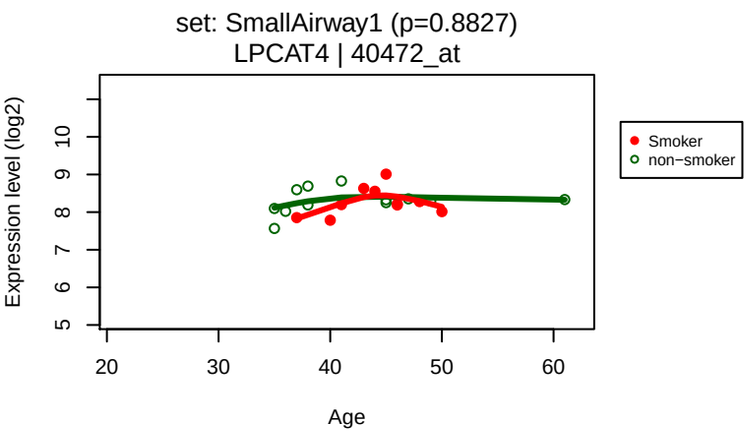
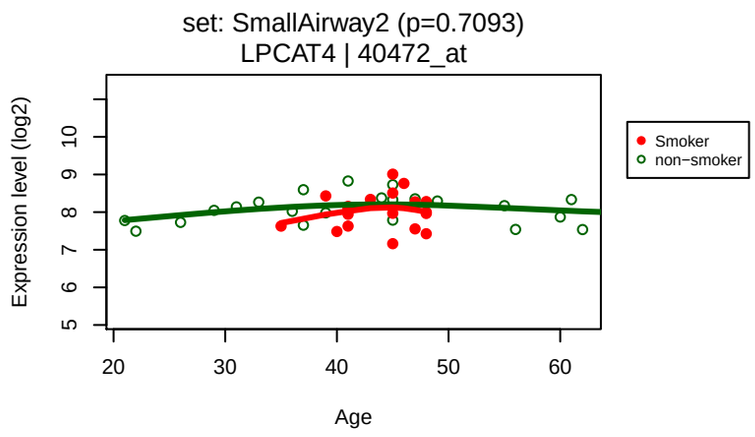
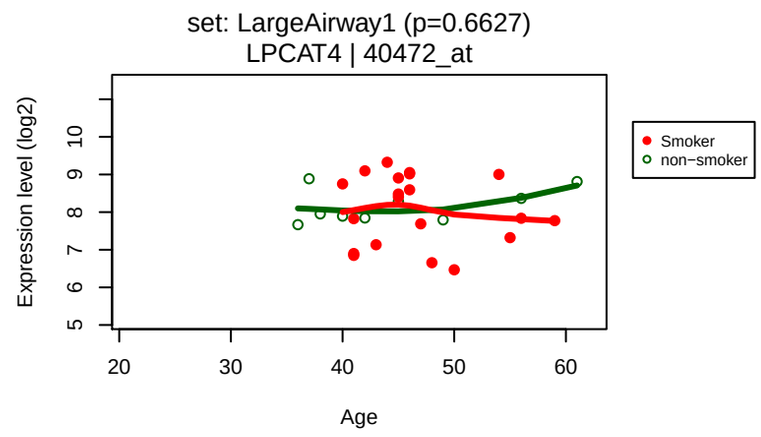

Gene expression of smokers and nonsmokers
gene: LPCAT4 | 40472_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8827
p=0.7093
p=0.039
p=0.6627
p=0.0376
p=1e−04
p=0.6123
p=0.0509
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

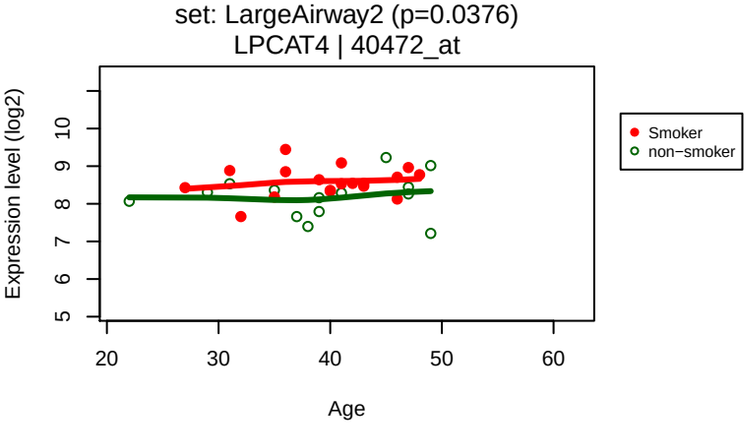

Gene expression ratio (smokers/nonsmokers)
gene: LPCAT4 | 40472_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8827
p=0.7093
p=0.039
p=0.6627
p=0.0376
p=1e−04
p=0.6123
p=0.0509
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

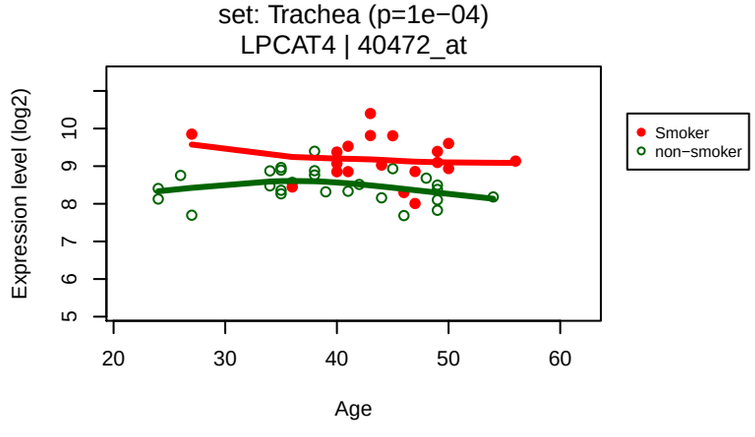

Gene expression ratio in Males
gene: LPCAT4 | 40472_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3233
p=0.022
p=0.0023
p=0.1362
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LPCAT4 | 40472_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2627
p=0.7633
p=0.0204
p=0.2942
SmallAirway2
LargeAirway2
Trachea
Macrophage2

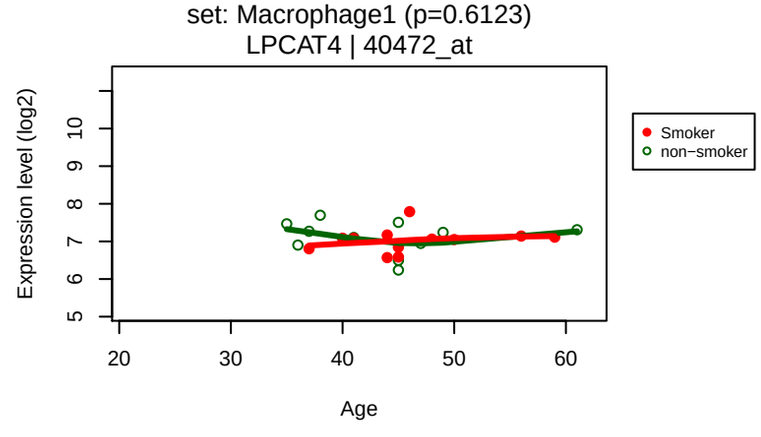

set: SmallAirway1 (p=0.8827)
LPCAT4 | 40472_at
Expression level (log2)
Age
Smoker
non−smoker

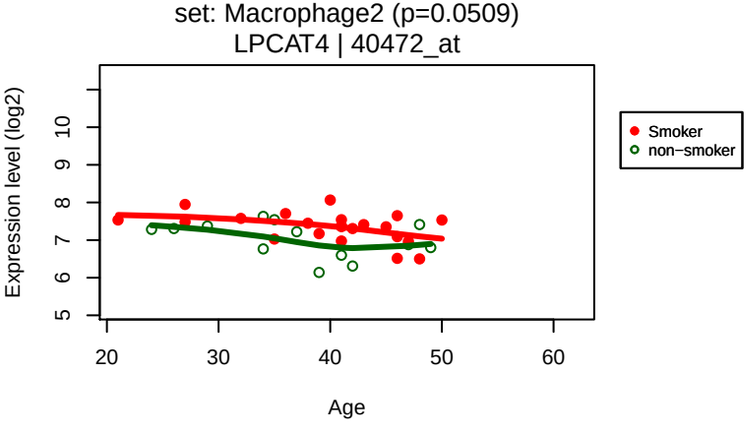

set: SmallAirway2 (p=0.7093)
LPCAT4 | 40472_at
Expression level (log2)
Age
Smoker
non−smoker

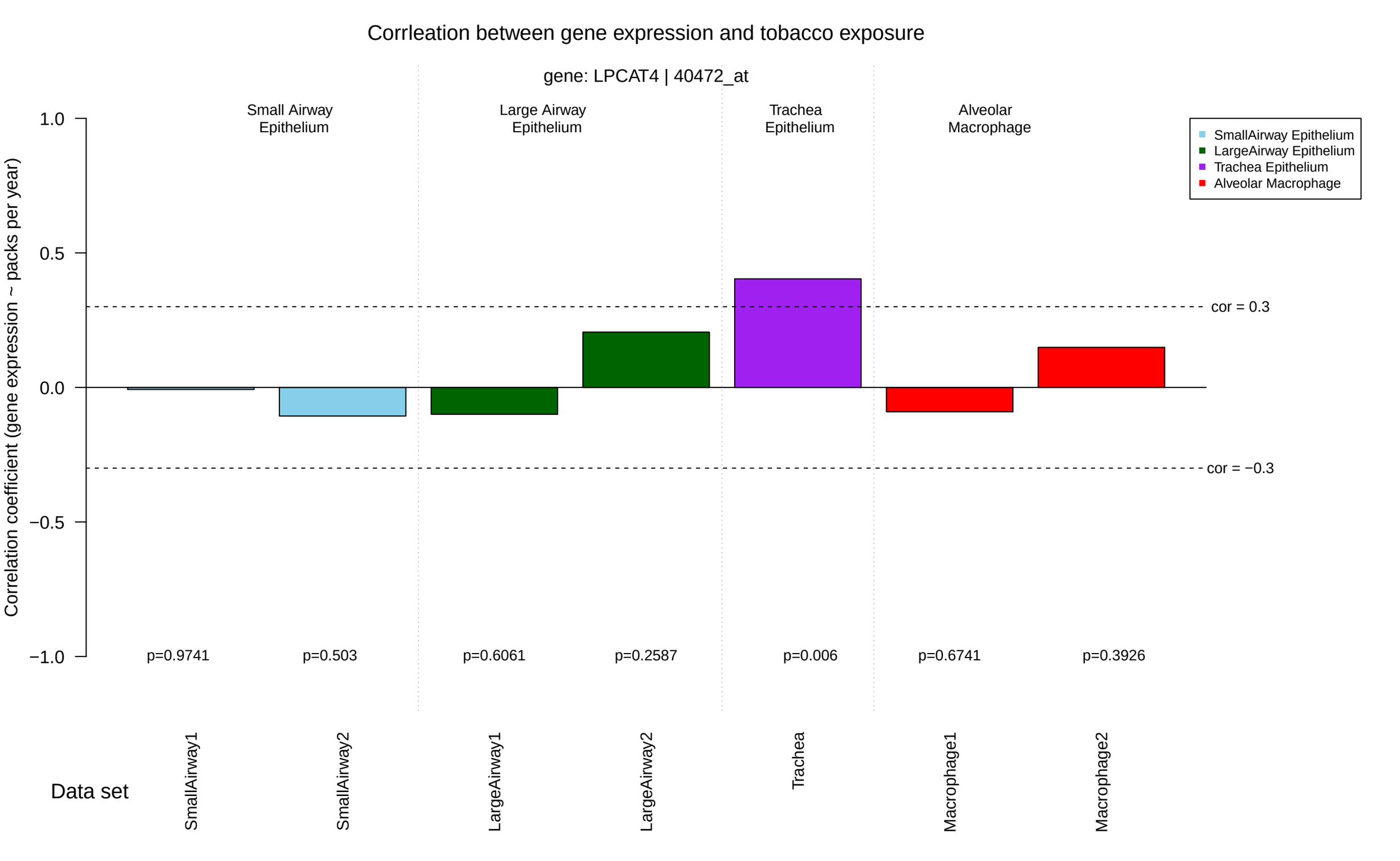

set: LargeAirway1 (p=0.6627)
LPCAT4 | 40472_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0376)
LPCAT4 | 40472_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
LPCAT4 | 40472_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6123)
LPCAT4 | 40472_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0509)
LPCAT4 | 40472_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LPCAT4 | 40472_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9741
p=0.503
p=0.6061
p=0.2587
p=0.006
p=0.6741
p=0.3926
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2